

MEMORANDUM FOR: THE RECORD

SUBJECT : Continuation of MKULTRA, Subproject No. 45

1. The scope of this subproject includes all those activities now engaged in by ~~the same personnel~~^B under the direction of ~~TSS/CD~~^{TSD/RB} with the exception of those cutout functions specifically mentioned in connection with other MKULTRA subprojects. In general, the research effort under this subproject will continue along the lines laid down in previous years. These involve the synthesis and pharmacological and clinical evaluation of compounds of those chemical families known to have application in the psychochemical and "K" fields. During the past year important progress has been made in the area related to stressor compounds and the relationship of these materials to the physiological pathways through which both stress and the reaction to it are mediated in human beings. As is indicated in the attached proposal, the work of the past year has progressed to the point where more definitive experiments on the stress reaction can be carried out. Primarily this was brought about by the characterization of several new materials which produce stress reaction in humans and the application of some new clinical methods of measuring the extent of the disturbance produced. During the next year proportionally more effort will be expended on the problem of the development of new

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"knock-out" types of agents since progress has been slower than is desirable in this direction and because a new approach to the problem has been worked out. //

2. ^C [REDACTED] also serves as a general consultant to the Agency, provides services of a sensitive nature on an ad hoc basis, and serves as a cut-out in procurement problems.

3. The total cost of this project for a period of one year will not exceed ^{40,500.00} \$71,500.00. Charges should be made against Allotment 0525-1009-4902.

4. ^C [REDACTED] has been requested to submit a summary accounting or a copy of the Fund's annual audit report for the sponsor's inspection. Also, it has been requested that any unexpended funds shall be returned to the Agency.

5. Title to any permanent equipment purchased by funds granted ^C [REDACTED] shall be retained by ^B [REDACTED], in lieu of higher overhead rates.

6. It was mutually agreed that documentation and accounting for travel expenses which are normally reimbursable by ^B [REDACTED] shall conform with the accepted practices of the Fund.

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1. The scope of this subproject includes all those activities now engaged in by [REDACTED] under the direction of TSD/RS with the exception of those cutout functions specifically mentioned in connection with other MKULTRA subprojects. In general, the research effort under this subproject will continue along the lines laid down in previous years. These involve the synthesis and pharmacological and clinical evaluation of compounds of those chemical families known to have application in the psychochemical and "K" fields. During the coming year it is planned to concentrate more directly on the more practical aspects of the "knockout" problem. Enough new potent substances have become available lately to make such a change in emphasis worthwhile. In connection with this change it should be noted that certain findings made in [REDACTED] project at [REDACTED] which cannot be further exploited at that facility will be pursued at [REDACTED] in the future. For this reason it may be necessary to supplement the findings of this subproject from time to time during the year due to increases of scope.

2. [REDACTED] also serves as a general consultant to the Agency, provides services of a sensitive nature on an ad hoc basis, and serves as a cutout in procurement problems.

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 Authority: 68-157-75
 Date: June 1977

E2 IMPDET; CL BY 187475

3. The total cost of this project for a period of one year will not exceed \$40,000.00. Charges should be made against Allotment 2125-1390-3902.

4. [REDACTED] has been requested to submit a summary accounting or a copy of the Fund's annual audit report for the sponsor's inspection. Also, it has been requested that any unexpended funds shall be returned to the Agency.

5. Title to any permanent equipment purchased by funds granted

[REDACTED] shall be retained by [REDACTED] B

B [REDACTED] in lieu of higher overhead rates.

6. It was mutually agreed that documentation and accounting for travel expenses which are normally reimbursable by [REDACTED] shall conform with the accepted practices of the Fund.

[REDACTED] A
Chief
TSD/Research Branch

APPROVED FOR OBLIGATION OF FUNDS:

[REDACTED] A
Research Director

Date _____

Attachment: Proposal and Budget

Distribution: Original only

DRAFT

24 January 1964

MEMORANDUM FOR : THE RECORD

SUBJECT : MKULTRA, Subproject 149

1. This subproject is being established for the purpose of supporting realistic tests of certain development items and delivery systems of interest to TSD/BB.

2. During the course of development it is sometimes found that certain very necessary experiments or tests are not suited to ordinary laboratory facilities. At the same time, it would be difficult if not impossible to conduct such tests as operational field tests. This project is designed to provide a capability and facilities to fill this intermediate requirement.

3. The activities under this subproject will be conducted by Mr. [REDACTED], an individual in the import and export business, in [REDACTED]. Mr. [REDACTED] holds a TOP SECRET Treasury Department clearance and a SECRET Agency approval. He is completely witting of the aims and goals of his activities.

4. Mr. [REDACTED] possesses unique facilities and personal abilities which makes him invaluable in this kind of testing operation. Mr. [REDACTED] because of his peculiar talents and

capabilities as well as his excellent connections with all of the local law enforcement agencies, will provide a unique and essential capability. Because Mr. [REDACTED] is no longer re-
sident of the [REDACTED] area, it is necessary that a suitable replacement be provided in order that a capability for continuance of our activities be maintained.

5. The estimated cost of the project is \$10,000.00 for a period of one year. Charges should be made against Allotment Number 4125-1390-3902. Reimbursement will be made for services rendered.

6. Accounting for funds advanced and any equipment under this subproject will be in accordance with accounting procedures established by the [REDACTED] *Administrative Staff* TS [REDACTED] A

7. A memorandum of agreement along lines established by previous audit recommendations in like situations will be executed.

[REDACTED]
[REDACTED] A
Chief
TSD/Biological Branch

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SUBJECT: Request for Support of Research on the Mechanism of Brain Concussion

1. This is a request for financial support for research on the mechanism of brain concussion for the period 1 Feb 1956 to 1 Feb 1957.
- B 2. The resonance-cavitation theory upon which this research is to be based has been presented in the proposal submitted to the [REDACTED] dated 27 March 1954.
3. The program as originally submitted estimated the duration of the program to be from three to five years requesting a total of \$72,109 for the initial year.
- B 4. At the request of the [REDACTED] a reduced budget was submitted.
- B 5. [REDACTED], amounting to \$24,925, was then awarded to the [REDACTED] to support this program from 1 Feb 1955 to 1 Feb 1956.
6. The progress made to date under the above contract can be summarized as follows:

A. RESEARCH FACILITIES

The following research facilities have been established for the investigation of the very diverse aspects of the problems being studied:

- B a. [REDACTED]
Total of 2500 square feet of laboratory and office space equipped with much of the diversified machinery and apparatus necessary for research in this field.
- B b. Blast Range
A blast range has been established at [REDACTED] located approximately [REDACTED] of the main laboratory. This area is owned by the [REDACTED] and is closed to the public. Three blast test-series have been run to date.
- B c. [REDACTED]
Arrangements have been made with the [REDACTED] for use of their human cadavers. A test area has been assigned for this

WARNING NOTICE

NO INTELLIGENCE
VALUES AND METHODS INVOLVED

B. PERSONNEL

Both full-time technical personnel and part-time professional research personnel have been acquired and indoctrinated relative to their specific function.

C. TECHNICAL PROGRESS

Following is the technical progress made under the current contract:

- a. Specialized instrumentation and numerous testing techniques have been developed to obtain the desired dynamic data.
 - b. Considerable data has now been obtained supporting the resonance-cavitation theory of brain concussion.
 - c. Preliminary acceleration threshold data has been obtained for a fluid-filled glass simulated skull.
 - d. Data has been obtained on the nature and the magnitude of pressure fluctuations within a glass simulated skull subject to either impact or sound waves propagated in air.
 - e. Initial studies have been made on the simulated glass skull attempting to establish the cavitation patterns for various types of impact.
7. The proposed method and program plan remain the same as stated in the original proposal, except for the temporary deletion of the immersion blast study.
 8. The current level of activity on this project can be indicated by the most recent billing to the [redacted] for the month of November, which amounted to \$4,034.61.
 9. In the interest of efficiency and economy it is requested that at least this level of activity be maintained for the coming year.

WARNING NOTICE

**SENSITIVE INTELLIGENCE
DATA AND METHODS INVOLVED**

10.0 POTENTIAL APPLICATIONS OF THE RESEARCH FINDINGS

- 10.1 Trotter, W. defines brain concussion as: "an essentially transient state due to head injury which is of instantaneous onset, manifests widespread symptoms of purely paralytic kind, does not as such comprise any evidence of structural cerebral injury, and is always followed by amnesia for the actual moment of the accident."
- 10.2 The implication of the underlined portion of the above statement is that if a technique were devised to induce brain concussion without giving either advance warning or causing external physical trauma, the person upon recovery would be unable to recall what had happened to him. Under these conditions the same technique of producing the concussion could be re-used many times without disclosure of its nature.
- 10.3 First, considering the possibilities of direct impact to the head or body, it should be possible from the findings of this research program to determine the following:
 - a. Optimum design of impacting devices.
 - b. Optimum points of impact on skull or body for the specific effects desired.
 - c. Intensity of the blow for the effect desired.
- 10.4 In regard to the potential impacting devices, there are certain design requisites that are apparent at this time:
 - a. The impact should be delivered without advance warning.
 - b. The area of impact and force distribution should be such that surface trauma does not occur.
 - c. The intensity of the impacting force and its duration should be such as to obtain the desired effect.
 - d. The device should be as small and as silent as possible.
- 10.5 The specific impacting devices might take the form of any of the following:
 - a. A pancake type black-jack giving a high peak impact force with a low unit surface pressure.
 - b. Concealed or camouflaged spring-loaded impacting devices that trigger upon contact with the head.

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c. A projectile type impactor such as an air gun using a small shot filled sack for a projectile.

d. An explosive pad detonated in contact with the head or the body.

10.6 Let us now consider the possibilities of exciting the resonance cavitation directly without impact. There is considerable evidence that resonance cavitation can be induced directly in the following ways:

a. A blast wave propagated in air. (Blast Concussion)

b. Physical excitation with a mechanical driver or horn, tuned to the resonant frequency of the head.

10.7 A single blast pressure wave propagated in air must have considerable intensity in order to produce brain concussion, however, there is considerable evidence (Carver & Dinsley) that modification of the pressure wave can produce profound effects.

10.8 Excitation of the resonance cavitation by using a tuned driver at this time appears to be well within the realm of possibility. The neurotic-like manifestations normally associated with blast concussion could possibly be induced by this method. Use of this method, however, would require actual physical contact with the drivers.

10.9 Excitation of the resonance cavitation by tuned sound waves also appears to be a reasonable possibility. Concentration of the sound-field at some remote point could be effected with acoustical lenses and reflectors. The blast duration would be in the order of a tenth of a second. Masking of a noise of this duration should not be too difficult.

11.0 It would possibly be advantageous to establish the effectiveness of both of the above methods as a tool in brain-wash therapy. A full knowledge of the method and the resulting sequela should be of aid to any person forced to submit to such treatment.

12.0 Possibly the most significant potential aspect of this study would be in the development of practical means of giving a person immunity, even though temporary, to brain concussion. One technique that appears to have potentialities involves the introduction of a small quantity of gas, approximately 1 cc, into the spinal cord. This gas bubble would then normally migrate to the ventricles located at the centrum of the brain. The ability of this bubble to expand under dynamic loading would be most effective in preventing resonance cavitation from occurring.

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MATERIAL FOR THE RECORD

MKSEARCH, OFTEN/CHICKWIT

MKSEARCH was the name given to the continuation of the MKULTRA program. Funding commenced in FY 1966, and ended in FY 1972. Its purpose was to develop, test, and evaluate capabilities in the covert use of biological, chemical, and radioactive material systems and techniques for producing predictable human behavioral and/or physiological changes in support of highly sensitive operational requirements.

OFTEN/CHICKWIT

In 1967 the Office of Research and Development (ORD) and the Edgewood Arsenal Research Laboratories undertook a program for doing research on the identification and characterization of drugs that could influence human behavior. Edgewood had the facilities for the full range of laboratory and clinical testing. A phased program was envisioned that would consist of acquisition of drugs and chemical compounds believed to have effects on the behavior of humans, and testing and evaluating these materials through laboratory procedures and toxicological studies. Compounds believed promising as a result of tests on animals were then to be evaluated clinically with human subjects at Edgewood. Substances of potential use would then be analyzed structurally as a basis for identifying and synthesizing possible new derivatives of greater utility.

The program was divided into two projects. Project OFTEN was to deal with testing the toxicological, transmissivity and behavioral effects of drugs in animals and, ultimately, humans. Project CHICKWIT was concerned with acquiring information on new drug developments in Europe and the Orient, and with acquiring samples.

There is a discrepancy between the testimony of DOD and CIA regarding the testing at Edgewood Arsenal in June 1973. While there is agreement that human testing occurred at that place and time, there is disagreement as to who was responsible for financing and sponsorship. (See hearings before the Subcommittee on Health and Scientific Research of the Senate Human Resources Committee, September 21, 1977.)

THE DIRECTOR OF CENTRAL INTELLIGENCE

WASHINGTON, D. C. 20505

Office of Legislative Counsel

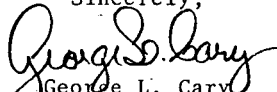
23 December 1977

Honorable Daniel K. Inouye, Chairman
Select Committee on Intelligence
United States Senate
Washington, D.C. 20510

Dear Mr. Chairman:

During Admiral Turner's 3 August 1977 testimony before your Committee and the Senate Human Resources Subcommittee on Health and Scientific Research, you asked whether any Agency employees had been terminated because of their participation in MKULTRA Subproject 3. Admiral Turner indicated he did not believe any employee had been terminated, but would have Agency records searched on this question. Our records have been searched and the results confirm the Director's testimony that no such actions were taken.

Sincerely,


George L. Cary
Legislative Counsel

QKHILLTOP DEFINITION

QKHILLTOP was a cryptonym assigned in 1954 to a project to study Chinese Communist brainwashing techniques and to develop interrogation techniques. Most of the early studies are believed to have been conducted by the Cornell University Medical School Human Ecology Study Programs. The effort was absorbed into the MKULTRA program and the QKHILLTOP cryptonym became obsolete. The Society for the investigation of Human Ecology, later the Human Ecology Fund, was an outgrowth of the QKHILLTOP.